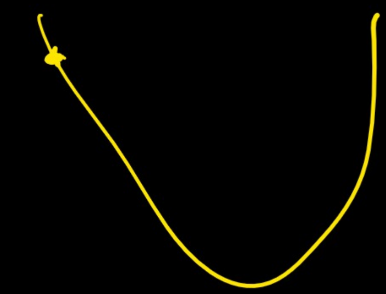
linear Regression — Calculations

x	y	$\hat{y}$
1	2	0
2	3	0
3	5	0
4	4	0
5	6	0
6	?	



$$y = mx + c$$

$$J(m, c) = \frac{1}{2n} \sum (\hat{y} - y)^2$$

$$\frac{\partial J}{\partial m} = \frac{1}{n} \sum x(\hat{y} - y)$$

$$\frac{\partial J}{\partial c} = \frac{1}{n} \sum (\hat{y} - y)$$

$$m_n = m_o - \eta \frac{\partial J}{\partial m}$$

$$c_n = c_o - \eta \frac{\partial J}{\partial c}$$

$$y = mx + c$$

$m = 0$

$c = 0$

$\eta = 0.05$

$n = 5$

$\hat{y} = 0 \cdot \cancel{x} \; 0 = 0$

$$\frac{\partial J}{\partial m} = \frac{1}{n} \sum x(\hat{y} - y)$$

$$= \frac{1}{5}\left(1(0-2) + 2\times(0-3) + 3\times(0-5) + 4(0-4) + 5(0-6)\right)$$

$$= -13.8$$

$$m_n = 0 - 0.05 \times (-13.8)$$

$$= 0.69$$

$$\frac{\partial J}{\partial c} = \frac{1}{n} \sum (\hat{y} - y)$$

$$= \frac{1}{5}\left((0-2) + (0-3) + (0-5) + (0-4) + (0-6)\right)$$

$$= -4$$

$$c_n = 0 - 0.05 \times -4$$

$= 0.20$

$m = 0.69, \quad c = 0.20$

$$\boxed{\hat{y} = 0.69x + 0.20}$$

x	y		$\hat{y}_2$
1	2		0.89
2	3		1.58
3	5		2.27
4	4		2.96
5	6		3.65

$$\frac{\partial J}{\partial m} = -5.61$$

$$\frac{\partial J}{\partial c} = -1.73$$

$$m_n = 0.69 - 0.05 \times -5.61 = 0.9705$$

$$c_n = 0.20 - 0.05 \times -1.73 = 0.2865$$

$$\hat{y} = 0.9705x + 0.2865$$

x	y	$\hat{y}_3$
1	2	1.2570
2	3	2.2275
3	5	3.1580
4	4	4.1185
5	6	5.1350

Evaluation metrics

MAE (mean absolute Error) = $\frac{1}{n} \sum |y_i - \hat{y}_i|$

on an average how much i am wrong ?

MSE (mean squared error) = $\frac{1}{n} \sum (y - \hat{y})^2$

RMSE (Root mean squared Error) = $\sqrt{MSE}$

(4) R^2 Score (Coefficient of Determination)

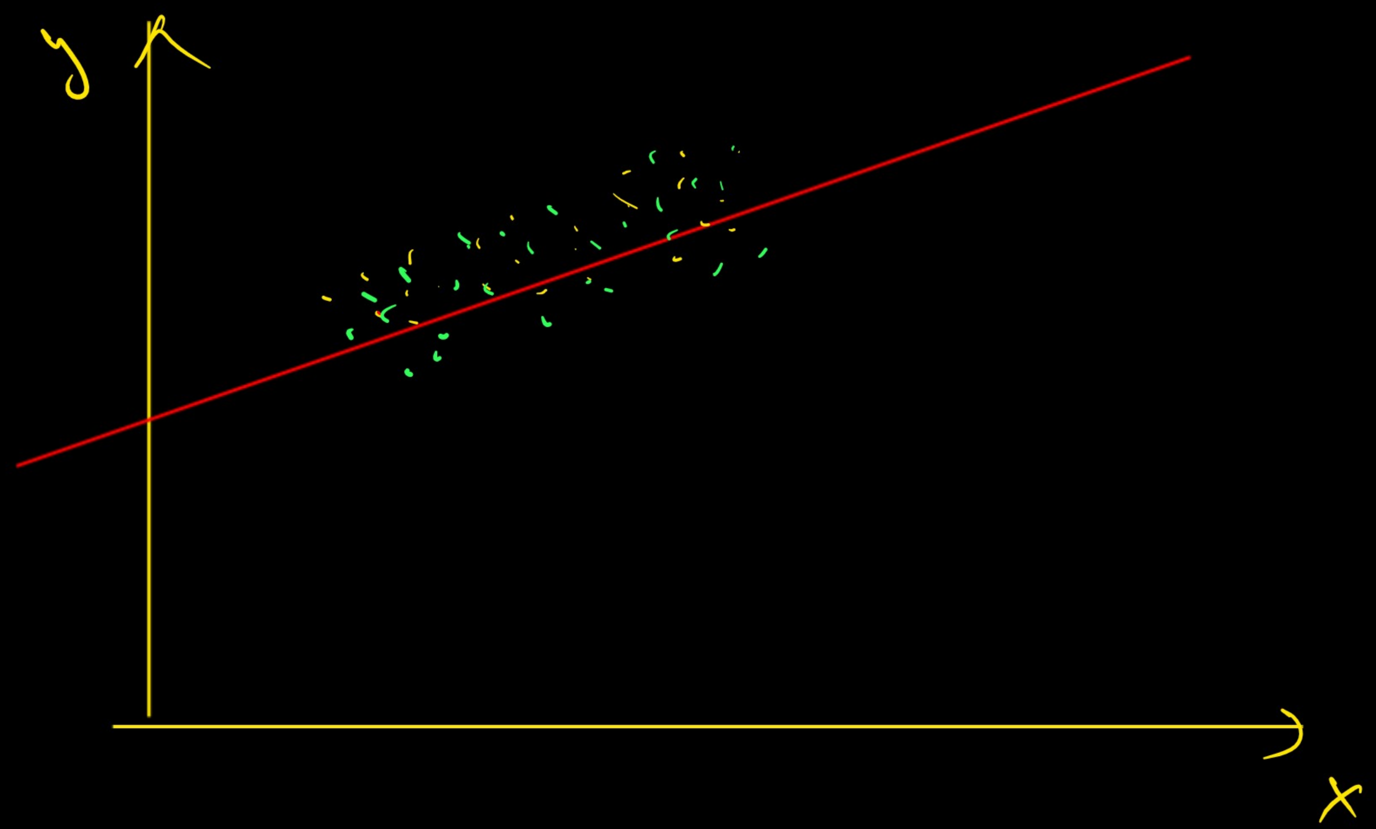
How well does my model explain the data.

$$R^2 = 1 - \frac{SS_{res}}{SS_{total}}$$

$R^2 \rightarrow 1.0 \rightarrow$ Perfect model

$R^2 \rightarrow 0 \Rightarrow$ model is not that great

$R^2 < 0 \Rightarrow$ It worse.



$$R^2 = 1 - \frac{SS_{res}}{SS_{total}}$$

$$SS_{res} = \sum (y - \hat{y})^2$$

$$SS_{tot} = \sum (y - \bar{y})^2$$

x	y	$\hat{y}$
1	10	11
2	12	13
3	14	14
4	16	15
5	18	17

$$SS_{res} = \Sigma (y - \hat{y})^2$$

$$= (10-11)^2 + (12-13)^2 + (14-14)^2 + (16-15)^2$$
$$+ (18-18)^2$$

$$= 4$$

$$SS_{res} = 4$$

$$SS_{total} = \Sigma (y - \bar{y})^2$$

$$= \bar{y} = \frac{10+12+14+16+18}{5} = 14$$

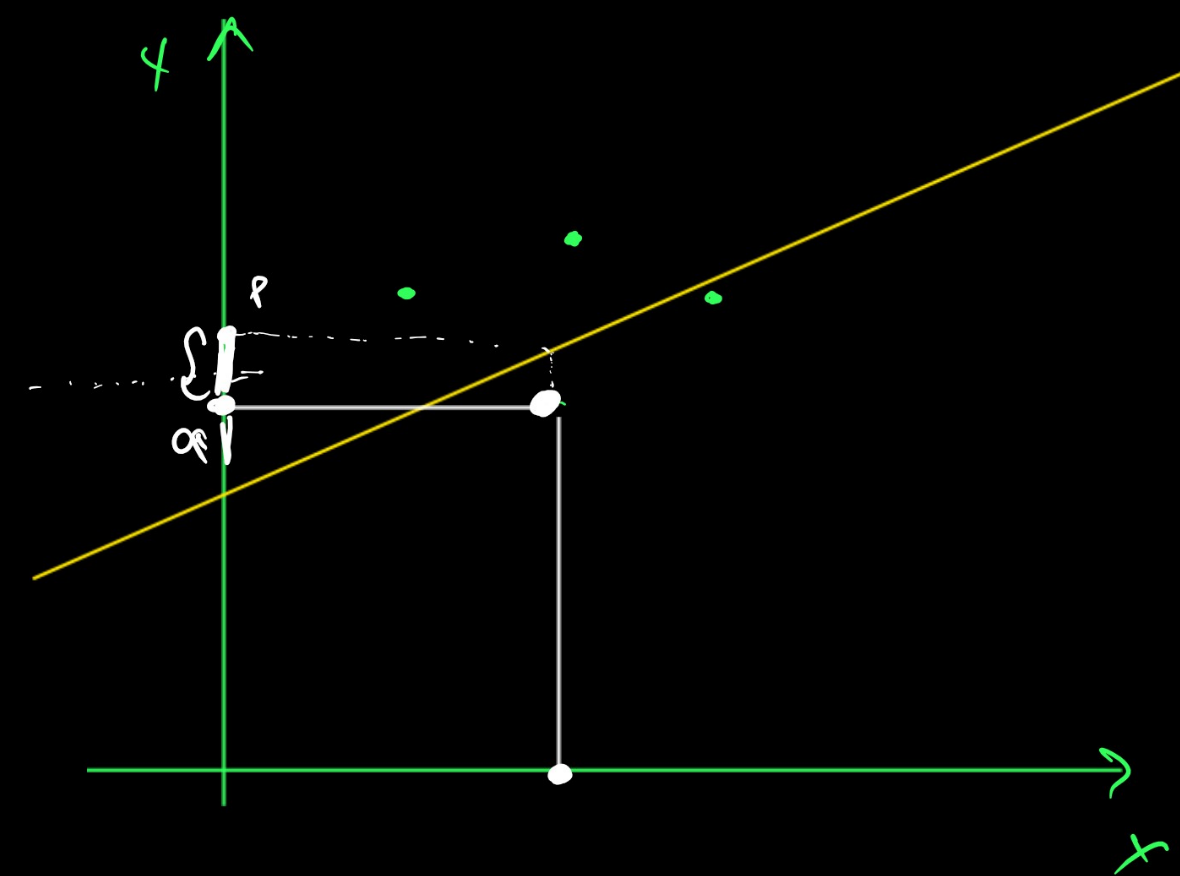
$$SS_{total} = (10-14)^2 + (12-14)^2 + (14-14)^2 + (16-14)^2 + (18-14)^2$$

$= 40$

$$R^2 = 1 - \frac{4}{40}$$

$= 1 - 0.1$

$= (0.9) \approx 1$



⑤ Adjusted $R^2 = 1 - \left(\frac{(1-R^2)(n-1)}{n-p-1} \right)$

n = number of Sample

P = number of features.